EEII II

Magnetic Load Attachment

Work Package Scope of Work — Engineering Design 2026

CONSTRAINT: No complete COTS electromagnets (COTS core material is permitted)

Main Work Elements

1. **Electromagnet design and simulation —** magnetic circuit design for the 3-point attachment — core geometry, air gap, turns and current — sized to meet the required holding force for the payload.
2. **Magnet manufacturing —** hand-wound coils for all 3 attachment points; COTS core material is permitted, but no pre-made electromagnets or solenoids may be procured.
3. **Drive electronics —** a switching circuit to energize and de-energize the magnet, including selection of its own COTS low-voltage supply.
4. **On-load indicator via current-sense —** characterizing the current signature difference between gripping a load and open air, and calibrating detection thresholds.
5. **Controller interface —** the isolated magnet PWM line and the current-sense signal, single-ended and capped below 2.048V, into the header already defined.
6. **Testing and validation —** confirming holding force under real load and validating the on-load indicator's reliability across repeated grip/release cycles.

Scoped for effort parity with the other four 2026 work packages (MCTR I, MCTR II, EEII I, EEII II, REII).